

19.1 部分习题参考简答

1. (1) 1; (2) 2π ; (3) 0; (4) $\frac{1}{2}$.

2. 提示: 分别在 $y \leq 0$, $0 < y < 1$ 和 $y \geq 1$ 时求 $F(y)$.

3. (1) $\frac{1}{2}$; (2) 0; (3) 0.

4. (1) $\frac{5}{2}xe^{-x^5} - (\frac{1}{2} + x)e^{-x^3} - \frac{1}{2x} \int_x^{x^2} e^{-xy^2} dy$;

(2) $\frac{b+2x}{x(b+x)} \sin x(b+x) - \frac{a+2x}{x(a+x)} \sin x(a+x)$;

(3) $\frac{2\ln(1+x^2)}{x}$;

(4) $\int_0^x (f_1'(x+u, x-u) - f_2'(x+u, x-u)) du + f(2x, 0)$;

(5) $x \left(\ln \frac{(1+5x^2)^4}{1+2x^2} \right) - 6x + 2\sqrt{1+x^2} \arctan \frac{x\sqrt{1+x^2}}{1+3x^2} + 4\sqrt{1+4x^2} \arctan \frac{x}{\sqrt{1+4x^2}}$;

(6) $2x \int_{x^2-x}^{x^2+x} \sin(x^2-s^2-x^4) ds + 2x \int_0^{x^2} \left(\int_{t-x}^{t+x} \cos(x^2-s^2-t^2) ds \right) dt$.

5. $3f(x) + 2xf'(x)$.

6. $F'(x) = 2x \int_0^x f(t, x^2) dt$.

7. $\int_0^{2\pi} tf(t \cos \theta, t \sin \theta) d\theta$.

9. (1) $\pi \ln \frac{1+a}{2}$. (2) 0.

10. (1) $\frac{\pi}{2} \ln(1+\sqrt{2})$; (2) $\arctan \frac{b-a}{ab+a+b}$.